NAMIT RATHOD

Houston, TX • +1-346-434-9556 • namitrathod07@gmail.com • linkedin.com/in/namit-rathod • github.com/namitrathod • Portfolio

EDUCATION

University of Houston - M.S. Computer Science *Aug 2024 – May 2026*

Coursework: Algorithms, Artificial Intelligence, Generative AI, Database Management Systems

EXPERIENCE

Software Engineer Intern | ScriptChain Health, California, US *Jan 2026 – Present*

- Architected a BFF pattern in FastAPI parsing FHIR medical records, reducing network JSON payloads by 80% and achieving O(1) rendering on Angular dashboards.
- Designed async Celery + Redis workflows for heavy patient dataset queries and EHR sync, ensuring high availability under concurrent clinical load.
- Integrated AWS DynamoDB, SES & S3 for scalable patient storage, email verification, and secure image handling; improved API response times by ~15% via query optimization.
- Built AI-driven food & exercise prediction APIs and exposed structured inference endpoints, delivering real-time health recommendations to frontend dashboards.
- Implemented OAuth 2.0 / JWT auth across distributed services; added Application Insights telemetry enabling A/B testing and 15% reduction in rendering latency.

Software Engineer | Cognizant, Bengaluru, India *Dec 2022 – Jul 2024*

- Led migration of a legacy monolithic UI to a React + TypeScript micro-frontend architecture, improving maintainability by 30% and enabling 4+ independent deployments per release cycle.
- Reduced average page interaction latency ~30% on a 1,000+ concurrent-user enterprise app by refactoring component state boundaries and eliminating excessive re-renders.
- Redesigned PostgreSQL schemas with advanced indexing + Redis caching, cutting query latency 30% and reducing data incidents 35% on systems handling millions of daily requests.
- Integrated OAuth 2.0 / JWT RBAC and enhanced CI/CD pipelines (GitHub Actions), reducing unauthorized access incidents 40% and deployment errors 20%.
- Architected reusable Fluent UI component library aligned with SOLID principles, cutting duplicate UI code 25% and accelerating cross-team delivery velocity.

Full-Stack Engineer Intern | Cognizant, Bengaluru, India *Jan 2022 – Nov 2022*

- Built a React admin dashboard with Node.js paginated APIs (PostgreSQL) adopted by 20+ support agents; optimized composite DB indexes, reducing query time from 3.2s → 300ms.
- Implemented RBAC enforcement, JWT auth with bcrypt, CI/CD validation (GitHub Actions), reducing regression defects 20% across release cycles.

PROJECTS

InboxSense - Semantic Email Intelligence | Next.js 14, Gemini 2.5, ChromaDB, RAG, Prisma, OAuth 2.0

- Engineered a RAG pipeline using gemini-embedding-2-preview to convert unstructured Gmail data into 3072-dim vectors for semantic search; reduced email noise by 70% via native Gmail query filters (-category:promotions -category:social).
- Built a context-aware AI Executive Assistant persona with Gemini 2.5 Flash using task-specific embedding prefixes; integrated Dockerized ChromaDB for localized, privacy-first vector storage with low-latency retrieval.
- Implemented resilient background sync engine with exponential backoff and 429-recovery; secured Gmail OAuth 2.0 via NextAuth.js with base64 MIME decoding; designed Prisma + SQLite schema for session and email metadata management.

Fair-KV Cache - LLM Multi-Tenancy | Python, FastAPI, Docker, Prometheus, OpenAI

- Designed per-tenant KV quota enforcement (oldest-first eviction + elastic borrowing) for multi-tenant LLM inference; reduced victim P95 latency under burst attacks while keeping evictions near zero.
- Shipped config-driven REST API (FastAPI) + Docker deploy; instrumented with Prometheus metrics and built a reproducible benchmark pipeline (LRU vs strict-quota vs FairKV), validating fairness with Jain's Index.

Distributed Job Processing System | Go, Kafka, Redis, Kubernetes, Prometheus, Grafana

- Built a Go-based distributed job processor handling 100K+ async tasks with Kafka priority scheduling, Redis caching (60% faster state lookup), retry/backoff, dead-letter queues, and K8s autoscaling.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), Java, Go, SQL

Frontend: React.js, Next.js, Angular, Redux Toolkit, Context API, Tailwind CSS, Fluent UI, RxJS

Backend: FastAPI, Node.js / Express, Spring Boot, REST APIs, GraphQL, OAuth 2.0 / JWT, Celery, Kafka

Databases & Cloud: PostgreSQL, MongoDB, Redis, DynamoDB, AWS (EC2/S3/SES), Azure, Docker, Kubernetes, CI/CD

AI / LLM: LLM API Integration, Prompt Engineering, Structured Output Validation, GPT-4o Vision, LangChain, RAG, Embeddings